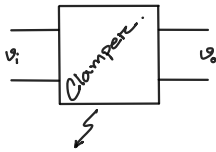
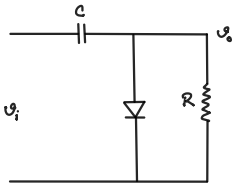
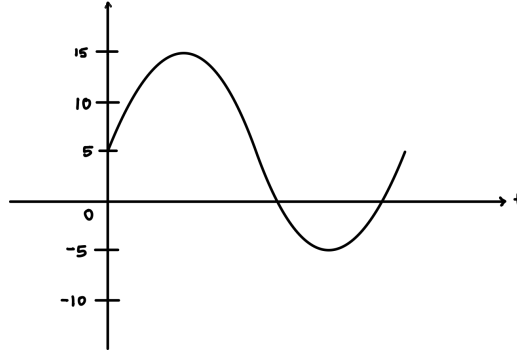
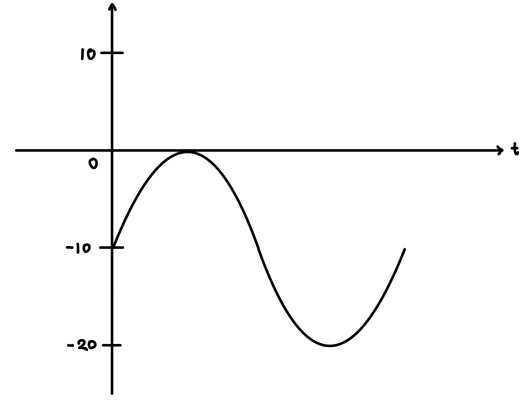
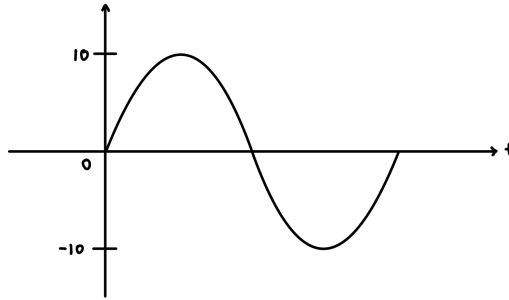


EEE205 (Tutorial 6)

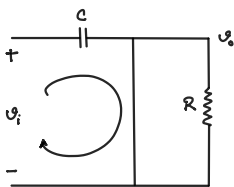
Clamper Circuits:



- Capacitor
- Diode
- DC Voltage Source
- Resistor



When diode is **ON**: At $0 < t < \frac{T}{2}$



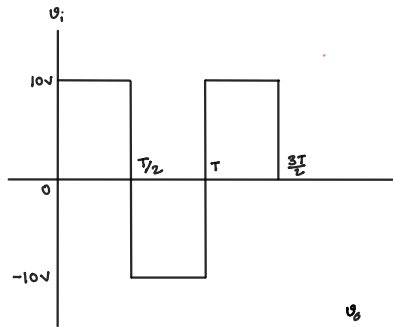
Capacitor being charged.

We know,

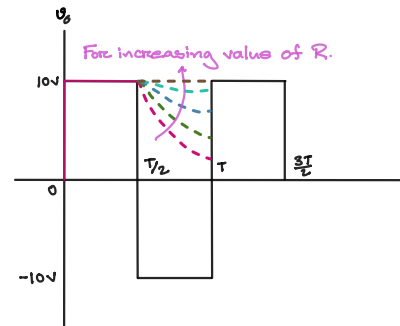
$$\tau = RC$$

$$\tau_1 = \tau_{DC} = 0$$

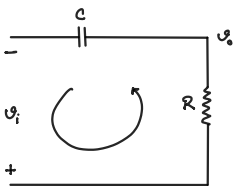
$\therefore$ Discharging is **Instantaneous**.



$$\therefore V_c = 10V$$



When diode is **OFF**: At $\frac{T}{2} < t < T$

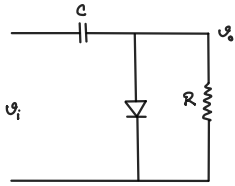


$$\therefore \tau_2 = RC \neq 0$$

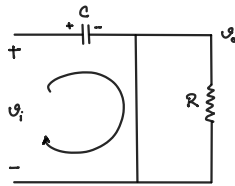
$\therefore R$ should be sufficiently larger

Capacitor being discharged.

Negative Clamping:

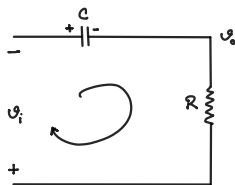


When diode is **ON**: At $0 < t < \frac{T}{2}$



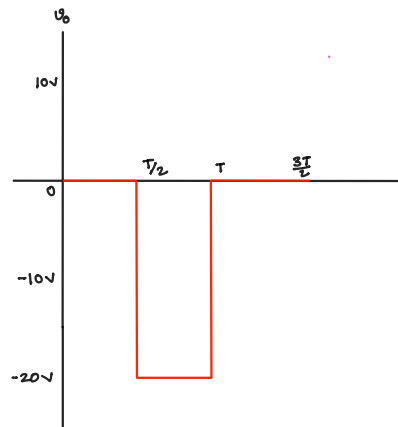
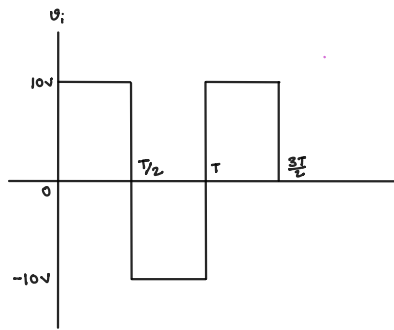
$$\therefore v_i - v_c = 0 \quad \therefore v_i = v_c = 10V \quad \text{and} \quad v_o = 0V$$

When diode is **OFF**: At $\frac{T}{2} < t < T$

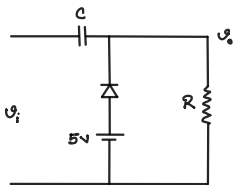


$$-v_i - v_c - v_o = 0$$

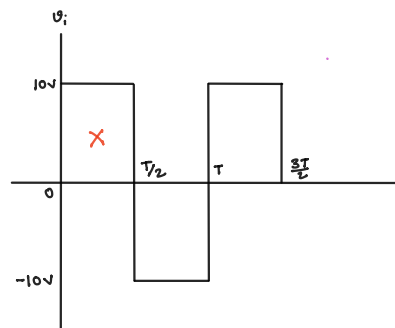
$$\Rightarrow v_o = -10 - 10 = -20V$$



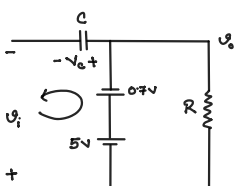
Positive Clamping:



For the half cycle, the diode appears to be **Reverse Biased**.



When the Diode is **ON**:

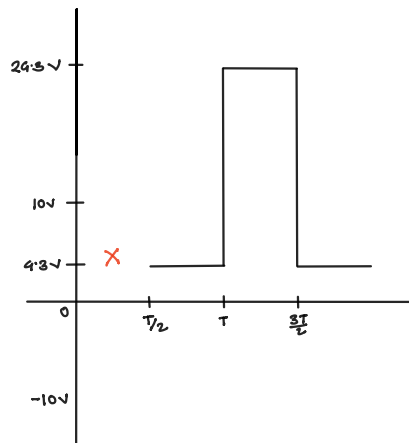


$$v_c = 0$$

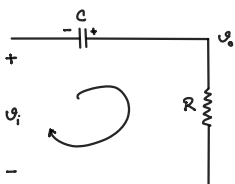
$$-v_i + v_c + 0.7 - 5 = 0$$

$$\Rightarrow -10 + v_c - 4.3 = 0$$

$$\therefore v_c = 14.3V \quad \text{and} \quad v_o = 4.3V$$



When the diode is **OFF**:



$$\therefore v_o = v_i + v_c = 10 + 14.3 = 24.3V$$